Finding Two Friction Coefficients from Energy Decay

A double pendulum, two unknowns, one linear fit

A double pendulum loses energy at a rate that is a *linear* function of the two friction coefficients. That one fact turns a hard nonlinear problem into an ordinary least-squares fit with two unknowns, so both coefficients come out of the same experiment, together.

1 Symbols

Angles θ_1 and θ_2 are measured from the downward vertical, each for its own rod. Everything in this list is already known from your measurements:

- m_1, m_2 — the masses of the upper and the lower rod.
- l_1 — distance from the top pivot to the centre of mass of the upper rod.
- l_2 — distance from the elbow to the centre of mass of the lower rod.
- l_3 — length of the upper rod, that is, pivot to elbow.
- I_1, I_2 — moment of inertia of each rod about its *own* centre of mass.

The length of the lower rod never appears: only the position of its centre of mass matters.

Five combinations appear again and again. Compute them once and the formulas stay short:

$$J_1 = I_1 + m_1 l_1^2 + m_2 l_3^2, \qquad J_2 = I_2 + m_2 l_2^2, \qquad J_{12} = m_2 l_3 l_2,$$
$$W_1 = (m_1 l_1 + m_2 l_3)\, g, \qquad W_2 = m_2 l_2\, g.$$

The two unknowns are the viscous friction coefficients: b_1 at the fixed top pivot and b_2 at the elbow, both in N·m·s. The friction torque at the top is $b_1\dot{\theta}_1$, because the upper rod turns against the frame. The friction torque at the elbow is $b_2(\dot{\theta}_2 - \dot{\theta}_1)$, because what matters there is how fast the two rods turn *relative to each other*.

2 Equations of motion

With viscous friction at both joints, the two Lagrange equations are

$$J_1\ddot{\theta}_1 + J_{12}\ddot{\theta}_2\cos(\theta_1-\theta_2) + J_{12}\dot{\theta}_2^2\sin(\theta_1-\theta_2) + W_1\sin\theta_1 = -b_1\dot{\theta}_1 \;+\; b_2(\dot{\theta}_2-\dot{\theta}_1),$$
$$J_2\ddot{\theta}_2 + J_{12}\ddot{\theta}_1\cos(\theta_1-\theta_2) - J_{12}\dot{\theta}_1^2\sin(\theta_1-\theta_2) + W_2\sin\theta_2 = -b_2(\dot{\theta}_2-\dot{\theta}_1).$$

Everything on the left is the ordinary frictionless double pendulum. Everything on the right is friction, and it carries both unknowns. Notice that b_2 appears in *both* equations, with opposite signs: that is Newton's third law at the elbow, and it is the reason the two coefficients cannot honestly be measured one at a time.

You will not have to solve these equations; they are here only to show where the friction enters. The method below uses energy instead.

3 Total energy

Kinetic energy, with the cross term that couples the two rods, and potential energy taken as zero when both rods hang straight down:

$$T = \tfrac{1}{2}J_1\dot{\theta}_1^2 + \tfrac{1}{2}J_2\dot{\theta}_2^2 + J_{12}\,\dot{\theta}_1\dot{\theta}_2\cos(\theta_1-\theta_2), \qquad V = W_1\big(1-\cos\theta_1\big) + W_2\big(1-\cos\theta_2\big).$$

The total energy is $E = T + V$. This is a quantity you can compute at every instant of your recording: you have θ_1 and θ_2 from the video, $\dot{\theta}_1$ and $\dot{\theta}_2$ by differentiating them, and all five constants from your measurements. **No unknowns enter E.** That is what makes the next step work.

4 The key identity

Multiply the first equation of motion by $\dot{\theta}_1$, the second by $\dot{\theta}_2$, and add them. Every conservative term collects into dE/dt, and what is left is the friction:

$$\frac{dE}{dt} \;=\; -b_1\dot{\theta}_1^2 \;-\; b_2\big(\dot{\theta}_2-\dot{\theta}_1\big)^2. \tag{1}$$

Read this carefully, because the whole method is in it. The energy always falls, since both squares are positive. The first term is the loss at the top pivot, the second is the loss at the elbow. And — this is the point — the right-hand side is a straight line in b_1 and b_2, with coefficients you can compute from the data. So it is a linear regression with two unknowns and **no intercept**: at rest nothing moves and nothing is lost, so the line must pass through the origin. You get one equation per video frame and only two unknowns, so the system is enormously overdetermined, which is exactly what you want.

Why both at once. If you fit b_1 alone you are forcing $b_2 = 0$, so b_1 quietly absorbs part of the elbow loss and comes out too large; fitting b_2 alone does the same in reverse. The two terms rise and fall on different schedules — the first is large when the whole pendulum swings together, the second when the rods fold against each other — and the regression uses exactly that difference to split the loss between the two joints.

5 Setting up the fit

Do **not** use dE/dt directly. Differentiating measured data amplifies noise, and E is already built from differentiated angles. Integrate (1) from the start of the record instead:

$$\underbrace{E(0)-E(t)}_{y(t)} \;=\; b_1 \underbrace{\int_0^t \dot{\theta}_1^2\,dt'}_{c_1(t)} + b_2 \underbrace{\int_0^t \left(\dot{\theta}_2-\dot{\theta}_1\right)^2 dt'}_{c_2(t)}. \tag{2}$$

It is the same straight line in b_1 and b_2, but now every column is a running integral, and integration smooths noise instead of amplifying it. The energy drop since the start is the left-hand side; c_1 and c_2 are the two columns of your design matrix. Smooth the angles lightly before differentiating — a filter window of a few percent of a period — then form y, c_1, c_2 frame by frame and solve the overdetermined system $y = b_1c_1 + b_2c_2$ by least squares.

The uncertainties a least-squares routine hands back are optimistic, because successive points of an integrated curve are not independent; and they say nothing about whether the model is right, as a precise fit to the wrong friction law still looks precise. Treat the spread of b_1 and b_2 across separate runs as the honest uncertainty.

6 Four checks before you believe the numbers

1. **Both coefficients must be positive.** A negative one is not a small friction — it is a mistake, almost always in a moment of inertia, in a distance to a centre of mass, or in a sign inside E.
2. **Look at the residual.** Plot $y-(b_1c_1+b_2c_2)$ against time. It should wander around zero with no shape. A smooth bow means the friction is not purely viscous; dry friction is the usual culprit. You can test that without leaving linear least squares — add two more columns, $\int|\dot{\theta}_1|\,dt$ and $\int|\dot{\theta}_2-\dot{\theta}_1|\,dt$, and fit four coefficients instead of two.
3. **Check that the two columns are distinguishable.** Compute the correlation coefficient between c_1 and c_2. If it is above about 0.99, the two integrals grow almost in lockstep and the fit cannot really tell b_1 from b_2: you will get a number, but moving one coefficient up and the other down changes the fit almost not at all. The condition number of the design matrix says the same thing — much above 30, be suspicious. This is the most common way the method fails, and it fails silently.
4. **Use a series of runs, never one.** A single recording fixes only one number, some combination of b_1 and b_2, and it cannot tell you whether the viscous model is true at all. This is the most important point in the whole method, so it gets its own section.

7 Designing the series of runs

Vary two things, independently.

Vary how hard the elbow works. Release the pendulum once with both rods swung the same way — in phase, so the elbow barely bends and c_1 dominates — and once with the lower rod folded back against the upper one — out of phase, so the elbow works hard and c_2 dominates. The useful measure of a run is the ratio of its two columns at the end of the record, $c_2(T)/c_1(T)$. A set of runs whose ratios are all similar carries the information of a single run, however different the videos look to the eye. A set whose ratios differ by a factor of several breaks the correlation in check 3, and only then does the fit separate b_1 from b_2.

Vary the amplitude: take both large-angle and small-angle runs. Record a group of small-angle releases, a few tens of degrees, where the pendulum swings and never flips, and a group of large-angle releases where the lower rod goes over the top. Fit each group on its own and compare.

If the friction really is viscous, the two groups must give the same b_1 and b_2. The coefficients are properties of the joints; they cannot depend on how hard you pushed. If the small-angle group returns a systematically larger b_1 than the large-angle group, the friction is not purely viscous — a constant dry-friction torque at a pivot behaves like a viscous coefficient that grows as the speed falls. If it returns a smaller one, something is being lost that grows faster than the square of the speed, air drag being the usual candidate.

Nothing inside a single amplitude range can reveal any of this, which is why the difference between the two groups — not the error bar of a single fit — is what you should quote as the uncertainty.

Then fit everything at once. Stack the design matrices of all the runs on top of each other, stack the energy drops the same way, and solve for the same two unknowns in one least-squares problem: the well-conditioned runs then carry the ones that are not. Report the result as two numbers with units and uncertainties, and state how many runs went into the fit, what amplitudes they covered, and what the correlation between the two columns was.